

Ain Shams University

Faculty of Computer and Information Sciences

Internal Regulations for the Bachelor's Degree under the Credit-Hour System

General Rules

Faculty Departments

The Faculty consists of the following scientific departments:

1. Computer Science Department
2. Information Systems Department
3. Basic Sciences Department
4. Computer Systems Department
5. Scientific Computing Department

The Faculty may establish other departments in accordance with the provisions of the Universities Organization Law.

Academic Degrees

Ain Shams University, based on the recommendation of the Faculty Council of Computer and Information Sciences, awards the Bachelor's degree in Computer and Information Sciences in the following specializations:

1. Computer Science – Basic Program
2. Information Systems – Basic Program
3. Scientific Computing – Basic Program
4. Computer Systems – Basic Program

Duration of Study

The period of study at the Faculty of Computer and Information Sciences for obtaining the Bachelor's degree shall not be less than four years, provided that all requirements for obtaining the degree are completed.

Student Admission Requirements

Admission is open to holders of the General Secondary Education Certificate (Mathematics section) or its equivalent. Admission is based on grades, with geographical distribution taken into consideration according to the decisions of the Supreme Council of Universities, provided that the student has studied Physics and Mathematics II.

In addition, the following must be fulfilled:

- The medical examination must prove that the student is free from infectious diseases and is medically fit to pursue the study for which they are applying.
- A student may not be enrolled in more than one faculty at the same time.

Credit-Hour Study System

- Study is conducted under the credit-hour system. A credit hour is a unit used to measure the weight of a course and consists of one hour of theoretical lectures or two or three hours of practical study or applied training. The credit hour is the basis for calculating the student's semester and cumulative GPA.
- The academic year consists of two main semesters (a full academic year), and each semester lasts 15 weeks.
- Courses for each program are distributed among the academic semesters, and the number of study units (credit hours) assigned to each semester is determined according to the study plan.
- To graduate, a student must successfully complete 140 credit hours distributed across four academic levels.
- Courses are divided into compulsory courses, elective courses, Faculty requirements, and University requirements.
- Each course has its own course number and code.
- An elective course may not be a prerequisite for a compulsory course.
- Prerequisites for courses at different levels are determined according to objective academic principles based on subject and chronological sequencing. A student may not register for a course with a prerequisite until the prerequisite has been successfully completed.
- The student must determine their major specialization after successfully completing a total of 60 credit hours. The specialization may not be changed except with the approval of the Faculty Council.

Academic Levels

- Level One (Freshman): Newly enrolled student who has not yet completed 36 credit hours.
- Level Two (Sophomore): Student who has completed 36 credit hours.
- Level Three (Junior): Student who has completed 69 credit hours.
- Level Four (Senior): Student who has completed 104 credit hours.

Academic Calendar and Course Registration

The academic year is divided into two main semesters:

A – First Semester (Fall): Begins in September and lasts 15 weeks. Course registration takes place within three weeks before the beginning of the semester.

B – Second Semester (Spring): Begins in February and lasts 15 weeks. Course registration takes place within one week before the beginning of the semester.

A summer semester may also be offered:

C – Summer Semester: Begins on the first Saturday of July and lasts 7 weeks. Course registration takes place within one week before the beginning of the semester.

Registration Requirements

The maximum number of credit hours a student may register for in each semester is:

- Up to 21 credit hours for a student with a cumulative GPA of 3.00 or higher.
- Up to 18 credit hours for a student with a cumulative GPA of 2.00 or higher but less than 3.00.
- Up to 14 credit hours or 5 courses for a student with a cumulative GPA below 2.00.
- The maximum registration load in the first and second academic semesters is 21 credit hours. The Faculty Council may authorize an increase beyond the maximum registration limit for students who are close to graduation.

Academic Advising

- The Vice Dean for Education and Student Affairs, in consultation with department heads, appoints an academic advisor for each student to assist and guide them in applying the credit-hour system.
- The academic advisor meets with students, introduces them to the program, answers their questions, and guides them. Ultimately, the student is responsible for the courses in which they register according to their own choices.

The academic advisor is responsible for:

- A. Introducing students to all rules and regulations governing the program, especially new students, to ensure that they understand the regulations and procedures related to registration, course selection, prerequisites, examination rules, assessment, and other matters.
- B. Helping students plan their study program in a manner appropriate to their capacity, progress in the academic program, and results.
- C. Helping students select courses for each semester and explaining the concept of prerequisites.
- D. Following up on the student's progress during the semester.
- E. Solving students' problems during study in coordination with the program academic coordinator and Faculty administration when necessary.
- F. Providing advice, guidance, and academic assistance to students.
- G. Performing any other duties assigned by the Dean of the Faculty.

Degree Requirements

The Bachelor's degree requires 140 credit hours of courses according to the established program, with a cumulative GPA of at least 2.00, and successful completion of courses assessed on a Pass/Fail basis, which are not included in the cumulative GPA, such as summer training, seminars, and similar activities.

Students must attend at least 75% of the total actual hours (lectures, exercises, and practical sessions) for any course. Otherwise, the student will be prohibited from taking the examination and will be considered failed.

Summer training is a graduation requirement. The student must complete two summer training periods during their study. Each summer training period must last at least three weeks and is conducted according to the rules established by the program; it is an essential graduation requirement.

Before graduation, students must prepare a Bachelor's project. A student must have successfully completed at least 94 credit hours before registering for the graduation project. Each group of students selects a project topic in accordance with their interests and with the approval of the academic supervisor during the final academic level.

Summer Training

- Students must complete two summer training periods during their study. Each summer training period must last at least three weeks.
- The program academic advisor supervises and follows up on summer training.
- At the end of the training period, the student must submit a technical report to the academic advisor.
- Summer training is assessed as Pass/Fail. Each summer training period is equivalent to two credit hours and is not included in the cumulative GPA.

Examination and Student Assessment System

- Student work in each course is assessed through periodic examinations, scientific reports, discussions, and final examinations at the end of the semester.
- Each course has a written midterm examination and a written final examination. The study regulations determine examination grades, coursework grades, and any other assessments according to the nature of each course.
- Grades are distributed among the different assessment methods for each course. For a student to pass, they must obtain at least 60% of the total course grade and at least 30% of the written final examination grade.
- A student must attend at least 75% of lectures and practical and theoretical exercises to be allowed to take the final examination. If a student's absence in any course exceeds 25% without an accepted excuse, the Faculty Council may prevent the student from taking the final examination and the student receives a 0.0 grade in the course. If the student submits an excuse accepted by the Faculty Council, the course is recorded as "Withdrawn." A student is warned when their absence reaches 20%.
- A student fails if their total grade in the course is below 60%, or if they do not take the final written examination because they were prohibited from entering it due to excessive absence, cheating, or any other reason, or if they miss the final examination without an excuse accepted by the Faculty.
- The maximum grade for each course is 100 marks.

Grade Distribution for Courses with a Laboratory

Assessment	Percentage
Final Examination	50%
Midterm Examination	15%
Exercises and Participation	10%
Periodic Examinations	5%
Practical Examination	20%

Grade Distribution for Courses without a Laboratory

Assessment	Percentage
Final Examination	60%
Midterm Examination	15%
Exercises and Participation	20%
Periodic Examinations	5%

Course Grades

Course grades in the offered program are calculated as follows:

Grade	GPA	Percentage
A+	4.0	97% and above
A	4.0	93% to less than 97%
A-	3.7	89% to less than 93%
B+	3.3	84% to less than 89%
B	3.0	80% to less than 84%
B-	2.7	76% to less than 80%
C+	2.3	73% to less than 76%
C	2.0	70% to less than 73%
C-	1.7	67% to less than 70%
D+	1.3	64% to less than 67%
D	1.0	60% to less than 64%
F (Fail)	0	Less than 60%
I (Incomplete)	None	
P (Pass)	None	

Absence from the Written Examination

A student who misses the final examination of a course without an acceptable excuse approved by the Faculty Council is considered failed in the course and receives an F grade.

Academic Warning – Dismissal – Raising the Cumulative GPA

- A student receives an academic warning if their cumulative GPA falls below 2.00 in any semester, with the requirement to raise the cumulative GPA to at least 2.00.
- A student under academic warning is dismissed if their cumulative GPA remains below 2.00 for six main semesters.
- If the student does not meet graduation requirements within the maximum study period, which is twice the normal study duration according to the law, the student is dismissed.
- The Faculty Council may consider granting a student at risk of dismissal because they were unable to raise their cumulative GPA to at least 2.00 one final opportunity lasting two academic semesters to raise the cumulative GPA to 2.00 and complete graduation requirements, provided that the student has successfully completed at least 80% of the required credit hours.
- A student may repeat courses previously passed for the purpose of improving the cumulative GPA. The course must be retaken and re-examined, and the grade obtained in the latest attempt is counted, subject to a maximum of five courses. If the improvement is for the purpose of lifting an academic warning or meeting graduation requirements, both grades are recorded in the academic record.
- When a student repeats a course in which they previously received an F, the grade obtained on repetition is counted up to a maximum of B. For calculating the cumulative GPA, only the latest grade is counted, while both grades are recorded in the student's academic record.

Regularity of Study

Students enrolled in the offered program under these regulations must comply with the following:

- Students may attend classes regularly only after paying the full tuition fees.
- Interruption of study is allowed only for an excuse accepted by the Faculty Council. If the interruption is due to illness, a medical certificate from an authorized university body must be submitted. If the student misses any scheduled examinations because of illness, a medical certificate from an authorized university body must also be submitted.
- The student must immediately notify the Faculty of any change in their correspondence address.

Transfer of Credit Hours

A student may transfer/credit courses studied between credit-hour programs within the Faculty after paying the registration fees (credit-hour fees) applicable to the program; the transferred hours are included in the student's cumulative GPA.

Suspension of Student Registration

The Faculty Council may suspend a student's registration for up to two consecutive or non-consecutive academic years during the student's study at the Faculty if the student submits an acceptable excuse preventing regular attendance.

Graduation Project

The student registers for the graduation project, which consists of 6 compulsory credit hours.

Proposed Study Plan

The study plan specifies the courses, the number of study hours allocated to lectures and practical exercises for each course, and the distribution of courses over the two academic semesters.

Level One Courses

Semester	Course Code	Course Name	Credit Hours	Prerequisites
First	CIS160	Introduction to Computer Sciences	3	-
First	BSC121	Physics I	3	-
First	BSC122	Calculus I	3	-
First	HUM110	English Language I	2	-
First	BSC123	Probability & Statistics	3	-
First	STU130	Selected University Topic (1)	2	-
First	STU140	Selected University Topic (2)	2	-
Second	CIS150	Structured Programming	3	CIS160 – Introduction to Computer Sciences
Second	CIS124	Electronics	4	BSC121 – Physics I
Second	BSC125	Calculus II	3	BSC122 – Calculus I
Second	STU150	Selected University Topic (3)	2	-
Second	BSC126	Physics II	3	BSC121 – Physics I
Second	HUM119	Human Rights & Combating Corruption	2	-

Level Two Courses

Semester	Course Code	Course Name	Credit Hours	Prerequisites
Third	HUM113	Report Writing	2	-
Third	CIS250	Object-Oriented Programming	3	CIS150 – Structured Programming
Third	CIS270	Data Structure	3	CIS250 – Object-Oriented Programming
Third	BSC221	Discrete Mathematics	3	-
Third	CIS260	Logic Design	3	BSC121 – Physics I
Third	CIS280	Database Management Systems	3	CIS150 – Structured Programming
Third	CIS243	Artificial Intelligence	3	CIS150 – Structured Programming
Third	CIS240	Statistical Analysis	3	BSC123 – Probability & Statistics
Fourth	CIS220	Computer Organization & Architecture	3	CIS260 – Logic Design
Fourth	BSC225	Linear Algebra	3	-
Fourth	CIS230	Operations Research	3	-

Published semester totals: Third semester: 17 credit hours. Fourth semester: 15 credit hours.

Level Three – Computer Science Specialization

Semester	Course Code	Course Name	Credit Hours	Prerequisites
Fifth	CIS353	Operating Systems	3	CIS220 – Computer Organization & Architecture
Fifth	CIS365	Computer Networks	3	CIS160 – Introduction to Computer Sciences
Fifth	CIS290	System Analysis & Design	3	CIS280 – Database Management Systems
Fifth	CSC330	Pattern Recognition	3	CIS150 – Structured Programming; BSC123 – Probability
Fifth	SCO311	Computer Graphics	3	CIS150 – Structured Programming; BSC225 – Linear Algebra
Sixth	CIS380	Software Engineering	3	CIS150 – Structured Programming
Sixth	CIS340	Analysis & Design of Algorithms	3	CIS150 – Structured Programming
Sixth	CSC340	Natural Language Processing	3	CIS150 – Structured Programming
Sixth	INF311	Data Mining	3	CIS280 – Database Management Systems
Sixth	CSC360	Compiler Theory	3	BSC221 – Discrete Mathematics
Sixth	CSC350	Concepts of Programming Languages	3	CIS250 – Object-Oriented Programming

Published semester totals: Fifth semester: 15 credit hours. Sixth semester: 18 credit hours.

Level Three – Information Systems Specialization

Semester	Course Code	Course Name	Credit Hours	Prerequisites
Fifth	CIS353	Operating Systems	3	CIS220 – Computer Organization & Architecture
Fifth	CIS365	Computer Networks	3	CIS160 – Introduction to Computer Sciences
Fifth	CIS290	System Analysis & Design	3	CIS280 – Database Management Systems
Fifth	INF321	Data Analytics	3	INF312 – Statistical Inference
Fifth	INF311	Data Mining	3	CIS280 – Database Management Systems
Sixth	CIS380	Software Engineering	3	CIS150 – Structured Programming
Sixth	INF322	Digital Multimedia	3	CIS150 – Structured Programming
Sixth	INF312	Statistical Inference	3	BSC123 – Probability and Statistics
Sixth	INF323	Data Security	3	CIS365 – Computer Networks
Sixth	STI310	Selected Topics (1)	3	-

Published semester totals: Fifth semester: 15 credit hours. Sixth semester: 18 credit hours.

Level Three – Scientific Computing Specialization

Semester	Course Code	Course Name	Credit Hours	Prerequisites
Fifth	CIS353	Operating Systems	3	CIS220 – Computer Organization & Architecture
Fifth	CIS365	Computer Networks	3	CIS160 – Introduction to Computer Sciences
Fifth	CIS290	System Analysis & Design	3	CIS280 – Database Management Systems
Fifth	SCO311	Computer Graphics	3	CIS150 – Structured Programming; BSC225 – Linear Algebra
Fifth	SCO313	Numerical Computing	3	CIS150 – Structured Programming; BSC225 – Linear Algebra
Sixth	CIS380	Software Engineering	3	CIS150 – Structured Programming
Sixth	CIS340	Analysis & Design of Algorithms	3	CIS150 – Structured Programming
Sixth	SCO322	Machine Learning	3	BSC123 – Probability and Statistics
Sixth	CSC350	Concepts of Programming Languages	3	CIS250 – Object-Oriented Programming
Sixth	SCO321	Digital Signal Processing	3	BSC125 – Calculus II
Sixth	CSY330	High Performance Computing	3	CIS353 – Operating Systems

Published semester totals: Fifth semester: 15 credit hours. Sixth semester: 18 credit hours.

Level Three – Computer Systems Specialization

Semester	Course Code	Course Name	Credit Hours	Prerequisites
Fifth	CIS353	Operating Systems	3	CIS220 – Computer Organization & Architecture
Fifth	CIS365	Computer Networks	3	CIS160 – Introduction to Computer Sciences
Fifth	CIS290	System Analysis & Design	3	CIS280 – Database Management Systems
Fifth	SCO321	Digital Signal Processing	3	BSC125 – Calculus II
Fifth	CSY340	Microprocessors & Interfacing	3	CIS220 – Computer Organization & Architecture
Sixth	CIS380	Software Engineering	3	CIS150 – Structured Programming
Sixth	CIS340	Analysis & Design of Algorithms	3	CIS150 – Structured Programming
Sixth	CSY350	Embedded System	3	CSY340 – Microprocessors & Interfacing
Sixth	CSY320	Data Communication	3	BSC125 – Calculus II
Sixth	CSY330	High Performance Computing	3	CIS353 – Operating Systems
Sixth	CSY360	Parallel and Distributed Architectures	3	CSY330 – High Performance Computing

Published semester totals: Fifth semester: 15 credit hours. Sixth semester: 18 credit hours.

Level Four – Computer Science Specialization

Semester	Course Code	Course Name	Credit Hours	Prerequisites
Seventh	SCO436	Computer Animation	3	SCO311 – Computer Graphics
Seventh	CSC410	Neural Networks and Deep Learning	3	CIS243 – Artificial Intelligence
Seventh	CSC425	Image Processing	3	CIS150 – Structured Programming
Seventh	STC310	Selected Topic (1)	3	-
Seventh	STC320	Selected Topic (2)	3	-
Seventh	PRO400	Project	3	-
Eighth	INF423	Cyber Security	3	CIS365 – Computer Networks
Eighth	CSC310	Distributed Computing	3	CIS353 – Operating Systems
Eighth	STC330	Selected Topic (3)	3	-
Eighth	STC340	Selected Topic (4)	3	-
Eighth	STC350	Selected Topic (5)	3	-
Eighth	PRO400	Project	3	-

Level Four – Information Systems Specialization

Semester	Course Code	Course Name	Credit Hours	Prerequisites
Seventh	INF411	Data Science I	3	INF312 – Statistical Inference
Seventh	INF412	Cloud Computing	3	CIS365 – Computer Networks
Seventh	INF413	Geographic Information Systems	3	CIS280 – Database Management Systems
Seventh	STI320	Selected Topic (2)	3	-
Seventh	STI330	Selected Topic (3)	3	-
Seventh	PRO400	Project	3	-
Eighth	INF421	Mobile Computing	3	CIS150 – Structured Programming; CIS365 – Computer Networks
Eighth	INF422	Social Media Analytics	3	INF312 – Statistical Inference
Eighth	INF423	Cyber Security	3	CIS365 – Computer Networks
Eighth	STI340	Selected Topic (4)	3	-
Eighth	STI350	Selected Topic (5)	3	-
Eighth	PRO400	Project	3	-

Level Four – Scientific Computing Specialization

Semester	Course Code	Course Name	Credit Hours	Prerequisites
Seventh	SCO422	Computational Geometry	3	SCO311 – Computer Graphics
Seventh	SCO411	Neural Networks & Deep Learning	3	BSC225 – Linear Algebra
Seventh	SCO421	Computer Vision	3	CIS243 – Artificial Intelligence
Seventh	STS310	Selected Topic (1)	3	-
Seventh	STS320	Selected Topic (2)	3	-
Seventh	PRO400	Project	3	-
Eighth	CSY410	Computer and Network Security	3	CIS365
Eighth	SCO312	Modeling & Simulation	3	-
Eighth	STS330	Selected Topic (3)	3	-
Eighth	STS340	Selected Topic (4)	3	-
Eighth	STS350	Selected Topic (5)	3	-
Eighth	PRO400	Project	3	-

Level Four – Computer Systems Specialization

Semester	Course Code	Course Name	Credit Hours	Prerequisites
Seventh	CSY410	Computer and Network Security	3	-
Seventh	INF412	Cloud Computing	3	CIS365 – Computer Networks
Seventh	SCO421	Computer Vision	3	CIS243 – Artificial Intelligence
Seventh	STY310	Selected Topic (1)	3	-
Seventh	STY320	Selected Topic (2)	3	-
Seventh	PRO400	Project	3	-
Eighth	CSY420	Real-time Systems	3	CSY350 – Embedded System
Eighth	INF421	Mobile Computing	3	CIS150 – Structured Programming; CIS365 – Computer Networks
Eighth	STY330	Selected Topic (3)	3	-
Eighth	STY340	Selected Topic (4)	3	-
Eighth	STY350	Selected Topic (5)	3	-
Eighth	PRO400	Project	3	-